

Statement of Purpose

Program: Master of Science in Electrical and Computer Engineering (Advanced Study Program)

Applicant: Yang Sen (Liam) Lin

In the evolving landscape of robotic autonomy, I have identified a fundamental trade-off: geometric rules provide reliability but become brittle in dynamic environments; conversely, Generative Vision-Language-Action (VLA) models offer adaptability but suffer from hallucinations and high computational latency. My objective is to resolve this tension by investigating the mathematical interfaces between rigid control theory and probabilistic foundation models. I specifically chose the **CMU MS ECE - Advanced Study Program** for its Project Option, which will allow me to integrate the robust optimization frameworks of **Prof. Zico Kolter** with the generalist robot learning of **Prof. Deepak Pathak**. My goal is to architect “Trustworthy Embodied Systems” that constrain the creativity of VLMs with the provable safety guarantees of control theory.

My journey toward this objective began with a focus on algorithmic decoupling, a process that presented the most significant academic challenge of my undergraduate career. In my research accepted to **CoRL 2025** (“See, Point, Fly”), I attempted to formulate a geometric projection algorithm to ground 2D VLM prompts into 3D metric controls. Initially, my approach failed. The drone, while perfect in simulation, repeatedly crashed in real-world tests due to a severe Sim2Real gap—my pin-hole camera assumptions were too brittle to handle the noise of physical actuators. This failure forced me to navigate through a rigorous debugging process, pivoting from end-to-end generation to a “spatial grounding” approach. By applying vector calculus to decompose displacement vectors into velocity primitives, I eventually achieved a **93.9%** success rate in unstructured environments. However, this experience taught me a crucial lesson: static geometric priors are insufficient for non-linear, dynamic obstacles. To overcome this “reactivity bottleneck,” I realized I must evolve from building ad-hoc heuristics to developing **Generalist Learning Policies**.

Simultaneously, my research philosophy is defined by the constraints of the physical layer. While many researchers focus solely on algorithmic performance, my internship at **Wolley Inc.** grounded me in the realities of hardware architecture. There, I engineered C-based diagnostic tools to profile **CXL memory bandwidth** and host pressure. This experience gave me the low-level architectural intuition needed to optimize I/O efficiency for physical AI, distinguishing my approach as a full-stack system builder. I plan to extend this hardware-software co-design mindset during my upcoming internship at **OMRON SINIC X**, where I will distill Robotics Foundation Models for edge deployment. I recognized early on that algorithmic distillation is insufficient without optimizing the entire stack—from memory protocols to control loops—making the **Hardware/Software Systems** focus at CMU the ideal academic home for my work.

My academic preparation is defined by the convergence of Mechanical Engineering and Computer Science. While ME grounded me in the physical first principles of system dynamics, my dual focus in CS equipped me with the algorithmic reasoning of modern AI. **I demonstrated the ability to synthesize these fields at a graduate level by securing an A+ in Graduate Computer Vision.** This practical application prepares me for the rigorous theoretical training of CMU’s ECE curriculum.

At CMU, I will leverage the Advanced Study Program to bridge the gap between control theory and mod-

ern AI. I plan to take **18-771 Linear Systems** and **10-725 Optimization for Machine Learning** to build a rigorous theoretical foundation for safety guarantees. Simultaneously, I will enroll in **16-831 Statistical Techniques in Robotics** to master uncertainty handling in dynamic environments. Most importantly, I intend to utilize the **36-unit Project Option (18-980)** to synthesize these fields. My technical roadmap maps directly to two specific research groups. I aim to contribute to **Prof. Zico Kolter’s lab**, where I hope to learn how to formalize heuristic safety constraints into provably stable optimization layers, preventing VLM hallucination in safety-critical tasks. Simultaneously, I seek to collaborate with **Prof. Deepak Pathak** to investigate how self-supervised world models can enable drones to adapt to non-linear dynamics in milliseconds, addressing the generalization limits I encountered in my CoRL work.

Beyond the laboratory, I carry an “optimization mindset” from code to community. As the **GDSC Chapter Lead**, I directed 24 members to bridge the gap between academic theory and industry practice, organizing workshops that impacted over 40 student developers. I aim to bring this leadership to the CMU community, fostering collaboration between ECE and Robotics cohorts.

My long-term goal is to lead the R&D of **Ubiquitous Physical AI**. The CMU MS ECE will transform me from a robotics engineer who builds working prototypes into a **Computational Systems Architect** capable of defining the safety guarantees required to deploy autonomous agents at a global scale.